

simeon djambov

LadHyX, Ecole Polytechnique
91128 Palaiseau cedex, FRANCE

e-mail: simeon.djambov@polytechnique.edu
website: sdjambov.github.io

research

- ❖ **postdoctoral associate** 2026–
Laboratoire d'Hydrodynamique de l'Ecole Polytechnique (LadHyX) Palaiseau, FRANCE
 - adviser: Etienne Jambon-Puillet;
- ❖ **doctoral assistant** 2021–2025
Laboratory of Fluid Mechanics and Instabilities, EPFL Lausanne, SWITZERLAND
 - thesis title: “Gravity-driven thin films: flow and dissolution;”
 - director: François Gallaire;
- ❖ **visiting researcher** Feb–Aug 2024
Laboratoire Matière et Systèmes Complexes Paris, FRANCE
 - host: Sylvain Courrech du Pont;
- ❖ **research intern** 2017–2021
 - “Fingering instability on a sphere” (F. Gallaire, LFMI, Feb–Jul 2021);
 - “Impact-induced cavitation” (C. Raufaste & F. Celestini, INPHYNI, Apr–Jul 2020);
 - “Electric double-layer dynamics” (A. Würger, LOMA, May–Jun 2019);
 - “Turbulent friction in 2D rough-walled flows” (H. Kellay, LOMA, May 2018–Apr 2019);
 - “Femtosecond laser spectroscopy” (J. Degert, LOMA, May–Jun 2017);

education

- ❖ **ph.d. in mechanics** 2021–2025
EPFL Lausanne, SWITZERLAND
- ❖ **m.sc. in physics** 2019–2021
Université Côte d’Azur Nice, FRANCE
- ❖ **b.sc. in physics** 2016–2019
Université de Bordeaux Bordeaux, FRANCE
- ❖ **highschool diploma** 2011–2016
French language Highschool “Alphonse de Lamartine” Sofia, BULGARIA

languages

bulgarian (native); french (fluent); english (proficient); german (limited); portuguese (notions);

interests

piano (ABRSM, Grade 8), choir (CHUL), F. M. Dostoevsky, pétanque, sudoku, sailing, swimming;

publications

- [7] S. Djambov, A. Marcotte, F. Gallaire, P. G. Ledda. “Thin-film drainage becomes singular at saddles,” *submitted in Phys. Rev. Lett.*
- [6] M. Chaigne, A. Maller, **S. Djambov**, M. Berhanu. “Dissolution rate of a soluble surface in a unidirectional turbulent flow,” *under review in J. Fluid Mech.*
- [5] S. Djambov, F. Gallaire. “Linear instability of weakly non-parallel film flows,” *under review in J. Fluid Mech.*
- [4] S. Djambov, F. Gallaire. “Analytical solutions to the gravity-driven rain-fed free-surface Stokes equations,” *C. R. Méc.* **354**, 451, 2026.
- [3] S. Eghbali, **S. Djambov**, F. Gallaire. “Stability of a liquid layer draining around a horizontal cylinder: Interplay of capillary and gravity forces,” *Phys. Rev. Fluids* **9**(6), 063903, 2024.
- [2] S. Djambov, F. Gallaire. “Boundary-layer approach to the linear karren instability,” *J. Fluid Mech.* **977**, R3, 2023.
- [1] A. Vilquin, J. Jagielka, **S. Djambov**, H. Herouard, P. Fischer, C.-H. Bruneau, P. Chakraborty, G. Gioia H. Kellay. “Asymptotic turbulent friction in 2D rough-walled flows,” *Sci. Adv.* **7**(5), eabc6234, 2021.

teaching

- ❖ **epfl**: ME-444 Hydrodynamics (F. Gallaire; 2023, 2025); ME-466 Instability (F. Gallaire; 2023, 2024); ME-446 Two-Phase Flows and Heat Transfer (F. Gallaire; 2022); PHYS-106(d) Thermodynamics (S. Bréchet; 2022); PHYS-101(d) Mechanics (P. Müllhaupt; 2021); co-supervision of two master’s students (2023, 2024);

grants

- ❖ **epfl doc.mobility**: “Condensation on soluble substrates: a path to impact-less linear karren formation;” six-month-long stay funded for nearly 20 kCHF;